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TORRY HARRIS

PoV

Connectivity to capability

The new economics of telco data

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Executive summary

Telcos sit atop one of the world's richest yet most underutilized resources: data. Every call, connection, and digital interaction generates insights with economic and societal value. As mobile data volumes surge, Communication Service Providers (CSPs) have a rare opening - to evolve from network enablers to data monetizers. The question is no longer whether to act, but how.

Global mobile data traffic reached a record 145 exabytes per month in 2024, with 5G contributing to 35% of the volume. IoT exchanges and HD streaming are compounding this growth. Yet telco revenues remain largely flat. The value has shifted - from bandwidth to behavioral intelligence. CSPs that structure and commercialize this intelligence can create new revenue streams while anchoring their relevance in digital ecosystems.

This PoV provides a pathway for CSPs to transition from connectivity providers to data monetizers, helping them create new value in the data economy.

Why the urgency

Market compression:
Data demand is exponential, but revenue per bit continues to decline.

Regulatory alignment:
Frameworks such as the EU Data Act and AI Act institutionalize secure, transparent data sharing, creating legitimacy for DaaS.

Technology readiness:
Open APIs, federated architectures, and data clean rooms now enable privacy-preserving data collaboration at scale.

Barriers to monetization

Despite the logic, execution has lagged. Telcos face structural drag - organizational, technical, and cultural. Each of these frictions delays monetization yet also points directly to the levers that unlock it.

Legacy infrastructure and data silos - Fragmented systems block unified analytics. Turning telemetry into usable products demands data integration across network, billing, and customer domains. The integration cost often outweighs perceived value - until leadership reframes it as a long-term strategic asset.

Capability gaps - Infrastructure expertise rarely translates to market-oriented productization. DaaS requires cross-functional fluency - engineering precision, data science acumen, and commercial storytelling. Few telcos combine all three.

Weak developer experience - Without robust documentation, APIs, and sandbox environments, developers default to hyperscalers. In the data economy, friction equals abandonment.

Regulatory complexity - Privacy, data residency, and consent laws remain fragmented. Compliance is not an overlay - it must be engineered into every data product.

Four stages of DaaS maturity

Data monetization is not a leap; it’s a staircase. Leading CSPs have followed a predictable sequence - from raw data to co-created ecosystems.

Data Brokerage - Selling exhaust data
Monetize historical, anonymized datasets for research, urban planning, or location-based insights. Revenue is incremental but foundational - it builds cataloging discipline and governance muscle.

API-First Access - Productizing real-time signals
Transition from data dumps to live APIs. Expose network and device intelligence through secure, monetized endpoints. Developers embed these signals directly into workflows, generating recurring usage-based income.

Insight-as-a-Service - Turning analytics into decisions
Apply AI/ML to derive predictive insights. These are not datasets - they are decision accelerators. CSPs move up the value chain, offering domain-specific foresight for industries such as retail, logistics, or finance.

Ecosystem Enablement - Orchestrating outcomes
At full maturity, telcos act as ecosystem orchestrators - combining their data with partners’ algorithms and domain knowledge to deliver turnkey vertical solutions. This is where scale meets stickiness.

Each stage demands new capabilities, operating models, and risk appetites - but also opens new economic logic.

Stage	Primary Role & Focus	Monetization Model	Key CIO Priorities & Investments	Organizational Impact
Stage 1: Data Brokerage	Monetize passive assets by selling historical, anonymized data sets.	One-off or subscription-based sale of static, anonymized datasets.	▪ Catalog and tag datasets for reuse. ▪ Establish privacy impact assessment processes. ▪ Appoint legal/commercial owners for external data use.	Data is treated as an IT or network by-product; minimal cross-functional collaboration.
Stage 2: API-First Access	Productize access to real-time telco signals for developers and partners.	Usage-based models: per-call pricing, tiered SLAs, or monthly active developer fees.	▪ Invest in API gateway, metering, and consumption analytics. ▪ Develop a self-service developer portal with robust SDKs. ▪ Build a dedicated developer relations (DevX) team.	A shift to platform thinking emerges, requiring new roles like API product managers and developer support teams.

Stage	Primary Role & Focus	Monetization Model	Key CIO Priorities & Investments	Organizational Impact
Stage 3: Insight-as-a-Service	Deliver premium, analytics-powered data products using AI/ML.	High-value, recurring subscriptions for predictive models and decisioning insights.	▪ Operationalize MLOps: model management, drift detection, and explainability. ▪ Invest in data science talent and ML pipelines. ▪ Co-create value stories with clients to prove ROI.	Data science becomes a core revenue-generating function; it requires cross-functional "insight squads" blending product, data, and marketing expertise.
Stage 4: Ecosystem Enablement	Act as an "Ecosystem Orchestrator," delivering end-to-end vertical solutions.	Solution-oriented models: platform fees, revenue sharing, or solution licensing bundles.	▪ Build data fusion and third-party integration capabilities. ▪ Develop vertical go-to-market (GTM) strategies and partnerships. ▪ Formalize joint innovation frameworks with partners.	Requires strategic partnership teams, vertical GTM specialists, and full P&L ownership for data-driven solutions.

Five CIO levers to operationalize DaaS

Operationalizing DaaS requires systemic change, not tactical pilots. Five levers define how CIOs can translate architecture into advantage.

1 Adopt federated data mesh as a strategic asset framework

CSPs can no longer afford the rigidity of monolithic data lakes that centralize control but paralyze agility. A federated data mesh enables domain teams - network, marketing, enterprise - to own and manage their data as products, governed by a shared policy and architecture. This model shifts the bottleneck from integration to innovation: teams can publish, consume, and iterate on data products independently. It preserves trust and compliance by embedding governance into the mesh, not imposing it from above. When implemented effectively, data becomes a shared corporate utility - discoverable, interoperable, and monetizable in real time. The result is a measurable increase in time-to-insight and partner readiness.

Pivot: The mesh is not a technical pattern - it's an organizational philosophy that aligns autonomy with accountability.

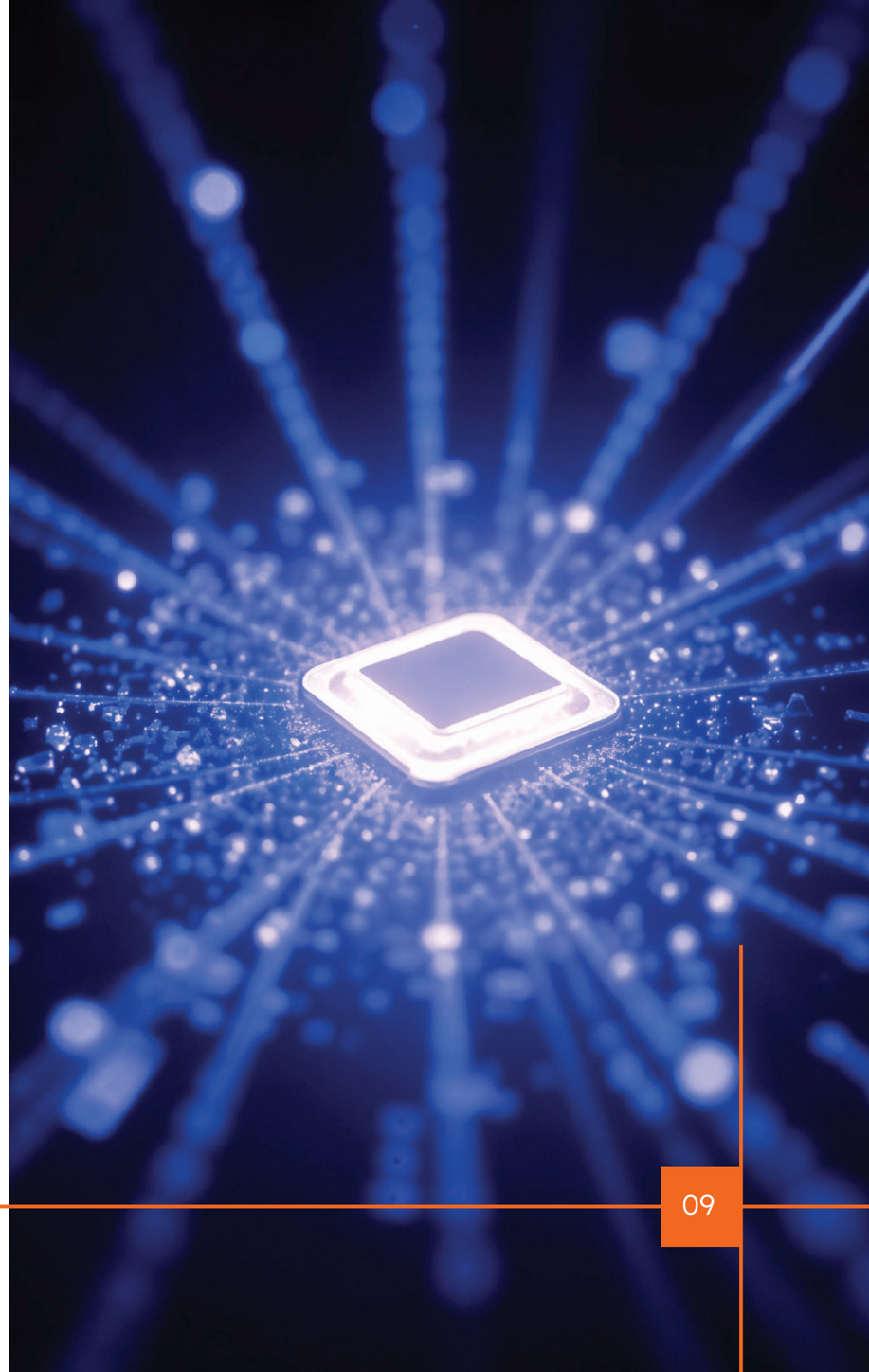
Vodafone's 'Data Ocean' initiative connects data domains across its European operations under a federated model. Each market curates its own data products - ranging from network telemetry to customer segmentwhile the Group Data Office enforces common metadata and privacy standards. This approach cut redundant pipelines by 30% and reduced time-to-insight from weeks to hours, creating the foundation for internal and external data monetization.

2 Monetize APIs with intent

Most telcos expose APIs as integration utilities; few treat them as products. Monetization requires intent - a lifecycle mindset that includes pricing, SLAs, developer experience, and commercial packaging.Programs like GSMA's Open Gateway - now spanning over 285 networks, nearly 80% of global mobile connections - illustrate how standardization can amplify reach and revenue assurance. The commercial logic is simple: usage drives monetization, and quality drives usage. Done right, APIs extend the telco's reach into ecosystems it doesn't own, while preserving brand trust and network-grade reliability.

Pivot: APIs become a new unit of growth - small in scope, massive in scale.

Telefónica's Koa platform operationalizes the GSMA Open Gateway framework across multiple regions. By offering APIs such as SIM Swap and Number Verification, Telefónica enables banks and fintechs to authenticate users in real time, reducing broker fraud by up to 70% through the implementation of CAMARA-standard APIs. The APIs are priced per transaction, turning network intelligence into a monetized digital product portfolio.



3 Institutionalize privacy by design

Privacy cannot be retrofitted - it must be architected from the start. Telcos handle vast amounts of personally identifiable data, and regulatory scrutiny is intensifying. Embedding privacy-by-design principles ensures that data monetization scales safely without compromising compliance or customer trust. Techniques such as differential privacy, homomorphic encryption, and secure multi-party computation allow value extraction from data without exposing identities. CSPs that lead with privacy can form high-trust partnerships with sectors like banking, healthcare, and government - where compliance risk is a barrier to entry. Moreover, privacy-by-design creates a competitive advantage: it transforms what others see as limitation into a mark of reliability.

Pivot: Trust, once engineered into data products, compounds like capital.

Deutsche Telekom deploys privacy-preserving data clean rooms to collaborate with ecosystem partners under strict GDPR compliance. In one initiative with a major European retailer, mobility data was anonymized and aggregated to optimize store locations and traffic flow analysis. The collaboration produced predictive insights without ever exposing raw user data - proof that privacy and performance can coexist.

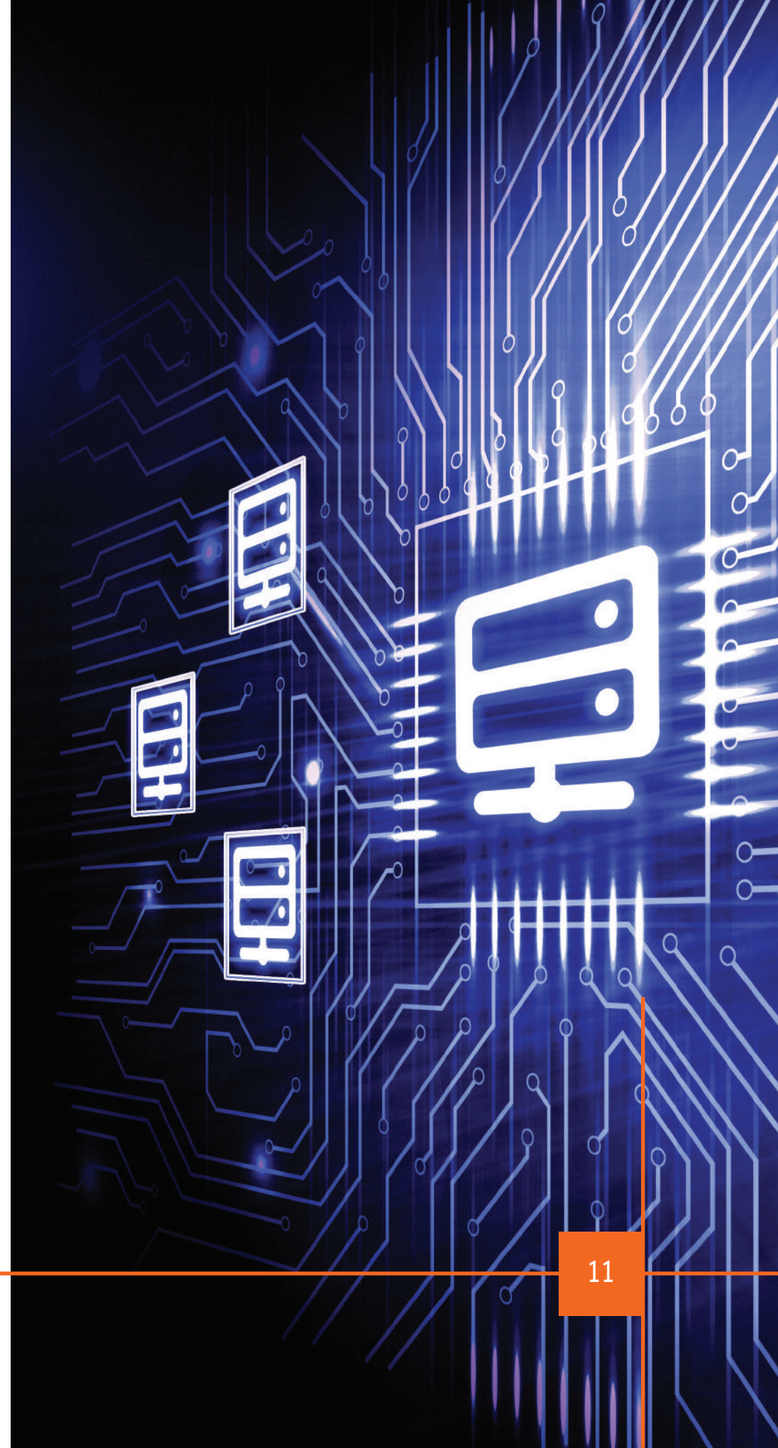
4 Build verticalized data marketplaces

Generic datasets rarely sustain long-term commercial value. Centralized data marketplaces provide discovery mechanisms, commercial structures, and operational support for DaaS offerings. Such platforms streamline procurement, accelerate onboarding, and reduce bespoke integration costs for enterprise customers. They also introduce standard contracts, pricing models, and usage policies - turning complex data buying into a repeatable transaction. When anchored in strong governance, these marketplaces become self-sustaining ecosystems that attract both data providers and consumers.

Over time, they evolve into a strategic GTM layer for telcos, amplifying reach across B2B verticals.

Pivot: Marketplaces don't just sell data - they sell confidence in how that data can be used.

Singtel's DataSpark platform exemplifies vertical marketplace design. By combining anonymized mobility and geospatial data, it provides real-time insights for Singapore's Land Transport Authority, helping optimize transit routes and urban planning. The platform has since expanded into tourism and retail analytics, standardizing how enterprises buy and integrate telco intelligence.



5 Establish a data product operating model (DPOM)

Data monetization cannot thrive in organizational ambiguity. A Data Product Operating Model establishes clear ownership, roles, and metrics around data commercialization. At its core is a Chief Data Product Officer (CDPO) who oversees cross-functional squads spanning data science, compliance, and go-to-market. These squads treat data as a P&L-managed product line, iterating on use cases, pricing, and value realization. The DPOM bridges technical stewardship with commercial accountability - linking architecture, governance, and business performance. When supported by agile governance and OKRs, it turns experimentation into execution discipline.

Pivot: Structure is the ultimate strategy - DPOMs make monetization measurable.

AT&T has institutionalized data operations under its Chief Data Office, embedding analytics and AI across the enterprise and treating network and IoT telemetry as business assets. It is actively commercializing these capabilities via APIs and data-products aimed at enterprises, turning network intelligence into monetized offerings. Cross-functional teams manage data-platform delivery, governance and monetization workflows. The result: faster time-to-value and a shift from connectivity to capability - evidenced by AT&T's IoT Console solution managing "1000+ APIs" across 250+ global carriers, enabling near-real-time device & connection oversight and accelerated service rollout.

Measuring what matters

To scale DaaS from isolated pilots to a core business pillar, CIOs must anchor success metrics and governance as foundational scaffolding.

Metric	Benchmark
Revenue Contribution	5-10% of total revenue within three years
Time-to-Market of Data Products	Fewer than 90 days from idea inception to product availability
API Adoption and Consumption	20% quarter-over-quarter increase in active API consumers (developers, partners), API calls, and subscriptions

Partnering for growth

Telecom growth will increasingly depend on how effectively operators turn data into value - not just for themselves, but for the industries they enable. Revenue from connectivity alone is finite; value creation now lies in the application of network intelligence across digital ecosystems.

That shift demands patience and pragmatism. Many CSPs are still reworking data foundations, rebalancing governance, and aligning business models around measurable outcomes. Progress comes unevenly - through pilots that expose structural gaps as much as they reveal potential.

Partnerships are central to this transition. The most credible gains emerge where telcos work with technology partners to co-develop operating frameworks, build interoperable architectures, and design monetization models that can withstand regulatory and commercial scrutiny. Each collaboration becomes a proving ground for how data can serve as infrastructure for broader economic activity.

Growth, in this context, is not about reinvention but reconfiguration - linking existing assets to new forms of demand through disciplined collaboration.

About us

Torry Harris Integration Solutions (THIS) is a global leader in business, technology, and IT consulting services with over 25 years of expertise in digital transformation. Specializing in digital ecosystem enablement, marketplace implementation, full life-cycle API management, digital integration, AI, and Data, Torry Harris empowers enterprises to unlock new growth opportunities and accelerate innovation. Today, the company is helping businesses worldwide establish Global Capability Centers (GCCs) and drive transformative, AI-powered initiatives that deliver measurable impact.

Headquartered in New Jersey, USA, THIS operates multiple offshore development centers in Bangalore, India, and has a strong global footprint with offices in Bristol and Slough (UK), Dubai (UAE), Riyadh (KSA), Dublin (Ireland), Munich (Germany), Paris (France), and Valencia (Spain).

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